HEURISTICS · PERMUTATION FLOW-SHOP PROBLEM

NEH, biased randomization and an adaptive β

RandNEH and ILS-ESP of Juan et al. (2014), built in Python with Numba and tested on five Taillard instances, from 20 to 500 jobs

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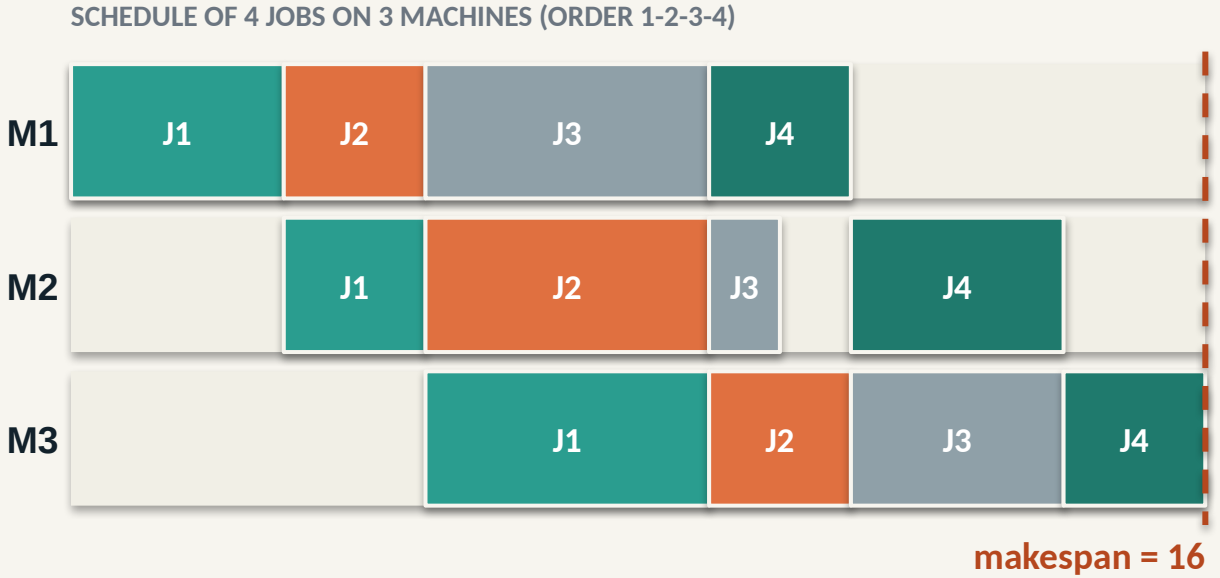
The permutation flow-shop problem

n jobs, m machines. Job j needs p[j,k] time units on machine k.

Every job visits the machines in the same order, and every machine processes the jobs in the same permutation.

Minimise the makespan: the time at which the last job leaves the last machine.

The search space is n! permutations.



Reordering the jobs shifts every block to the right or to the left, so the permutation decides the makespan.

C[i][k] = max(C[i−1][k], C[i][k−1]) + p[π(i), k]. Evaluating one permutation costs O(n·m). GAP % = 100 · (value − BKS) / BKS.

NEH: the constructive heuristic

TOTAL PROCESSING TIME OF JOB j

$$TPT(j) = \Sigma_k\, p[j, k]$$

1 Sort the jobs

by TPT, largest first

2 Take the next job

in that order, one at a time

3 Insert it

at the position with the lowest partial makespan

4 Repeat

until the n jobs are placed

Deterministic: one run, one solution. A good start, but it never explores alternatives.

Taillard's acceleration evaluates the k+1 insertion positions of a job in $O(k \cdot m)$ instead of $O(k^2 \cdot m)$, with forward (e), backward (q) and new-job (f) matrices. Ties go to the leftmost position.

Nawaz, Enscore and Ham (1983). Our implementation matches a naive NEH and reproduces the reference makespan of tai117 (26,797).

RandNEH: biased randomisation

Instead of always taking the first job of the TPT-sorted list, pick its position pos at random, biased towards the front. The whole list stays eligible: every remaining job has a probability, none is excluded.

Triangular (Juan et al.)

$pos = \lfloor r \cdot (1 - \sqrt{(1 - u)}) \rfloor$

No parameters. Probability falls linearly along the list.

Geometric (β)

pos ~ Geom(β), truncated to r

One parameter. A higher β concentrates the choice at the front.

Uniform (control)

$pos = \lfloor r \cdot u \rfloor$

No bias at all: a random order. It shows what the bias is worth.

MEAN CHOSEN POSITION WITH R = 100 JOBS LEFT (0 = THE FIRST OF THE LIST)

32.8

Triangular

9.0

Geo(0.1)

2.3

Geo(0.3)

49.5

Uniform

u ~ U(0,1), r = jobs still to place. The PDF leaves b undefined in X = b·(1 − √(1 − u)); we take b = r (assumption). Geo(β) keeps the same bias for any r.

The paper's RandNEH (Fig. 6)

RANDNEH(INSTANCE, DISTRIBUTION)

```
best  = NEH()                        // deterministic
tries = 0
while best is NEH and tries < n:
    tries += 1
    S = biased_construction()    // draw pos
                                 // for every job
    if makespan(S) < makespan(NEH):
        best = S
return best      // never worse than NEH
```

WHY A LOOP

One biased construction is not enough: in our first tests a single one did not beat NEH on average.

The loop keeps trying and stops at the first construction that wins, so the cost is the number of tries.

The output is never worse than NEH, and it differs from run to run.

Example, tai117: Geo(0.7) needs 2.4 constructions on average (53 ms).

Juan et al. (2014), Fig. 6. One construction = one full NEH pass with randomised job order. One run of the loop = one RandNEH solution.

ILS-ESP: the complete algorithm

ILS-ESP · EACH RUN IS AN INDEPENDENT AGENT

```
base = RandNEH();  base = local_search(base)
best = base
while time left:
    cur   = enhanced_swap(base)
    cur   = local_search(cur)
    delta = cost(cur) − cost(base)
    if delta < 0:                    // improvement
        base = cur;  last_imp = −delta
        was_imp = True;  update best
    elif 0 < delta ≤ last_imp and was_imp:
        base = cur;  was_imp = False   // Demon
return best
```

BUILDING BLOCKS

Enhanced swap. Swap two random positions, then move each of the two jobs to its best position.

Local search. Take each job out, in random order, and reinsert it at its best position; repeat until a full pass brings no gain.

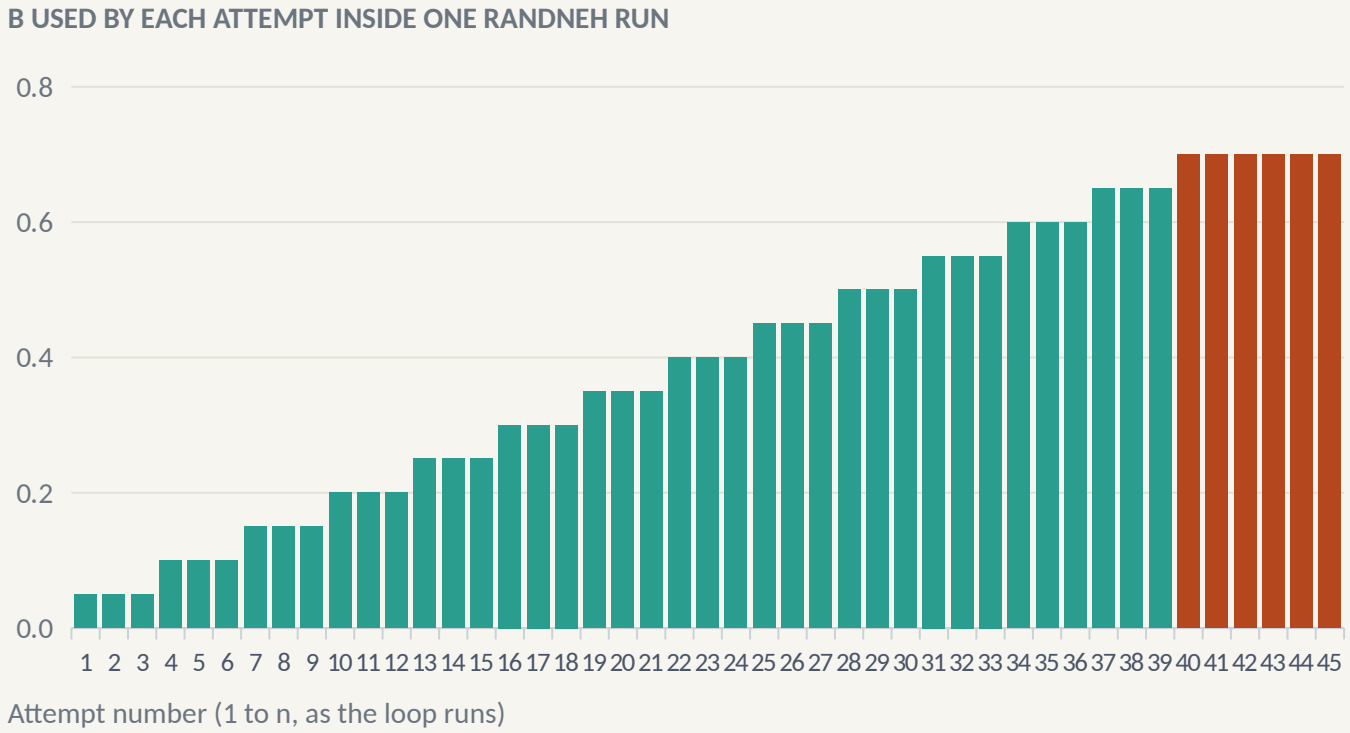
Demon acceptance. A worse solution is accepted only right after an improvement, and never by more than that improvement.

No parameters. It stops by time: 2 ms · n · m.

Juan et al. (2014). The slide's localSearch(baseSol) is corrected to currentSol. Parallel agents are independent seeds, emulated as the best of 10 runs.

Our proposal: an adaptive β

A fixed β has to be chosen in advance, and the best value depends on the instance. Geo-adapt starts with a wide selection and raises β every time the loop fails to beat NEH.



β FOR ATTEMPT t

$$\min(0.7,\ 0.05 + 0.05 \cdot \lfloor (t-1)/3 \rfloor)$$

Start at β = 0.05: a wide selection, very different constructions.

Every 3 failures, +0.05: the choice concentrates near NEH.

Cap at 0.7. A success ends the run; the next run restarts at 0.05.

The only signal used is whether a construction beat NEH. Parameters were fixed before the runs, not tuned.

Geo-adapt = the Fig. 6 loop with a different β at each attempt. Used as the constructive phase (Experiment A) and as the ILS-ESP start (Experiment B).

Experimental design

Five Taillard instances: tai001 (20×5), tai051 (50×20), tai081 (100×20), tai101 (200×20) and tai117 (500×20).

Experiment A · constructive phase

NEH against RandNEH (Fig. 6) with 9 distributions plus Geo-adapt: Tri, Geo(β) for β = 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, and Uniform.

30 runs per distribution and instance.

Experiment B · complete ILS-ESP

Only the start changes: NEH, Tri, Geo(0.1), Geo(0.3), Uniform and Geo-adapt, all with the Fig. 6 RandNEH.

10 seeds per start and instance, budget 2 ms · n · m, same time for all.

HOW WE MEASURE

GAP %

100 · (value − BKS) / BKS, the distance to the best known solution

Success rate

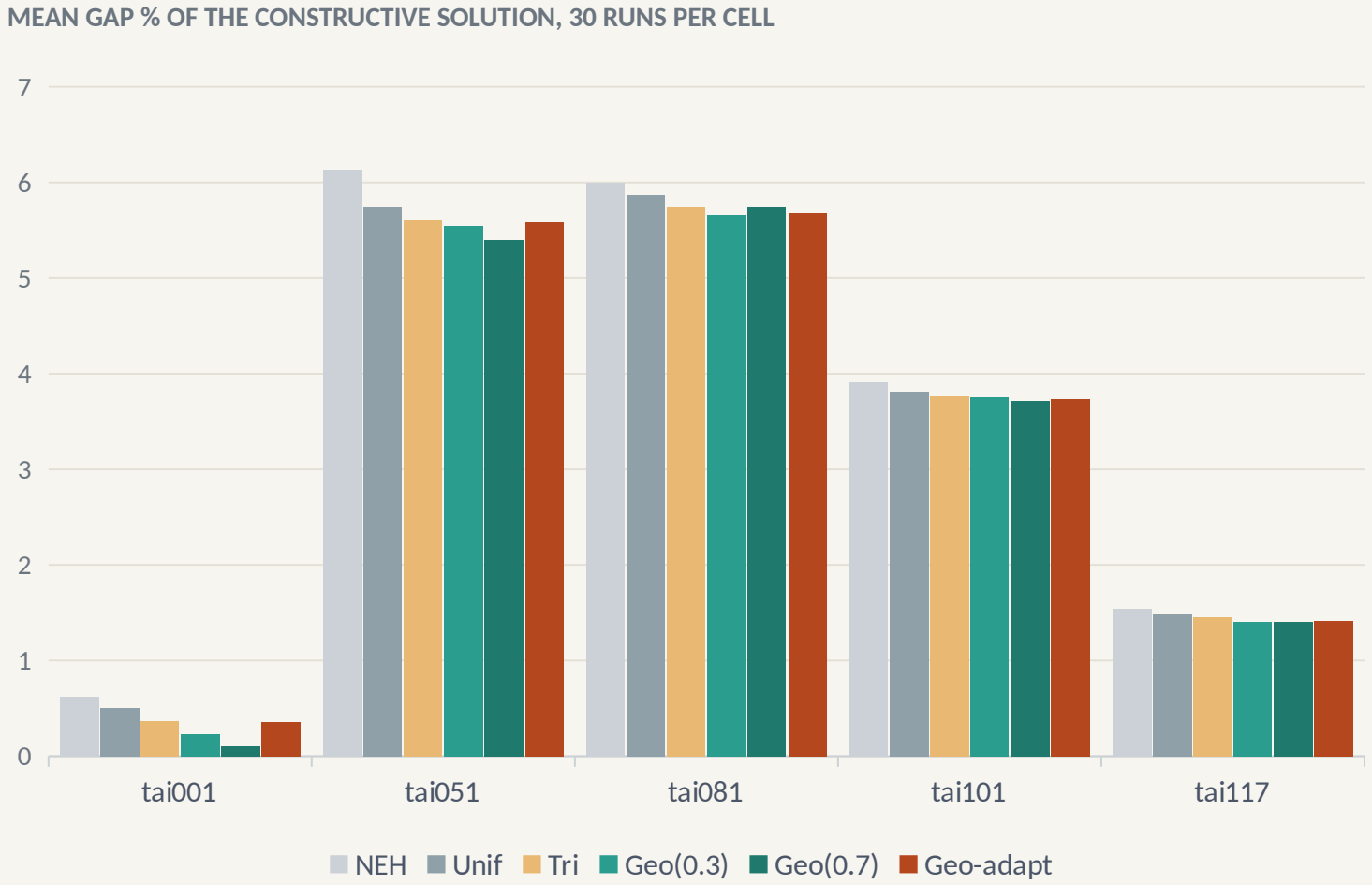
% of runs that beat NEH (Experiment A)

Cost

constructions and ms per run (A); iterations within the budget (B)

BKS: Taillard's table (tai001) and the Zenodo CSV (rest); only tai101 and tai117 are proven optimal. Mann-Whitney tests against NEH, uncorrected.

Every RandNEH beats NEH on average



100 %

of the runs beat NEH when n ≥ 50 (uniform: 67–97 % from tai081 on)

0.1 to 0.8

GAP points won over NEH: modest, because the loop stops at the first success

150 vs 2.4

constructions per run on tai117: uniform against Geo(0.7)

NEH is deterministic. tai001 (n = 20) allows only 20 attempts, which is why its gaps are the most uneven. Axis starts at 0.

Similar quality, very different cost

MEAN CONSTRUCTIONS PER RUN · AND MS PER RUN ON TAI117

Distribution	tai081	tai101	tai117	ms, tai117	GAP tai117
Tri	14.3	14.5	7.9	143	1.45
Geo(0.1)	8.3	10.6	2.7	60	1.40
Geo(0.3)	4.9	7.4	3.2	69	1.40
Geo(0.5)	5.1	6.9	2.6	53	1.39
Geo(0.7)	6.7	5.6	2.4	53	1.41
Unif	57.2	103.6	149.9	2429	1.49
Geo-adapt	**7.2**	**11.3**	**3.5**	**77**	**1.41**

Geo-adapt is neither cheaper nor better than a well-chosen fixed β. Its advantage is not having to choose it.

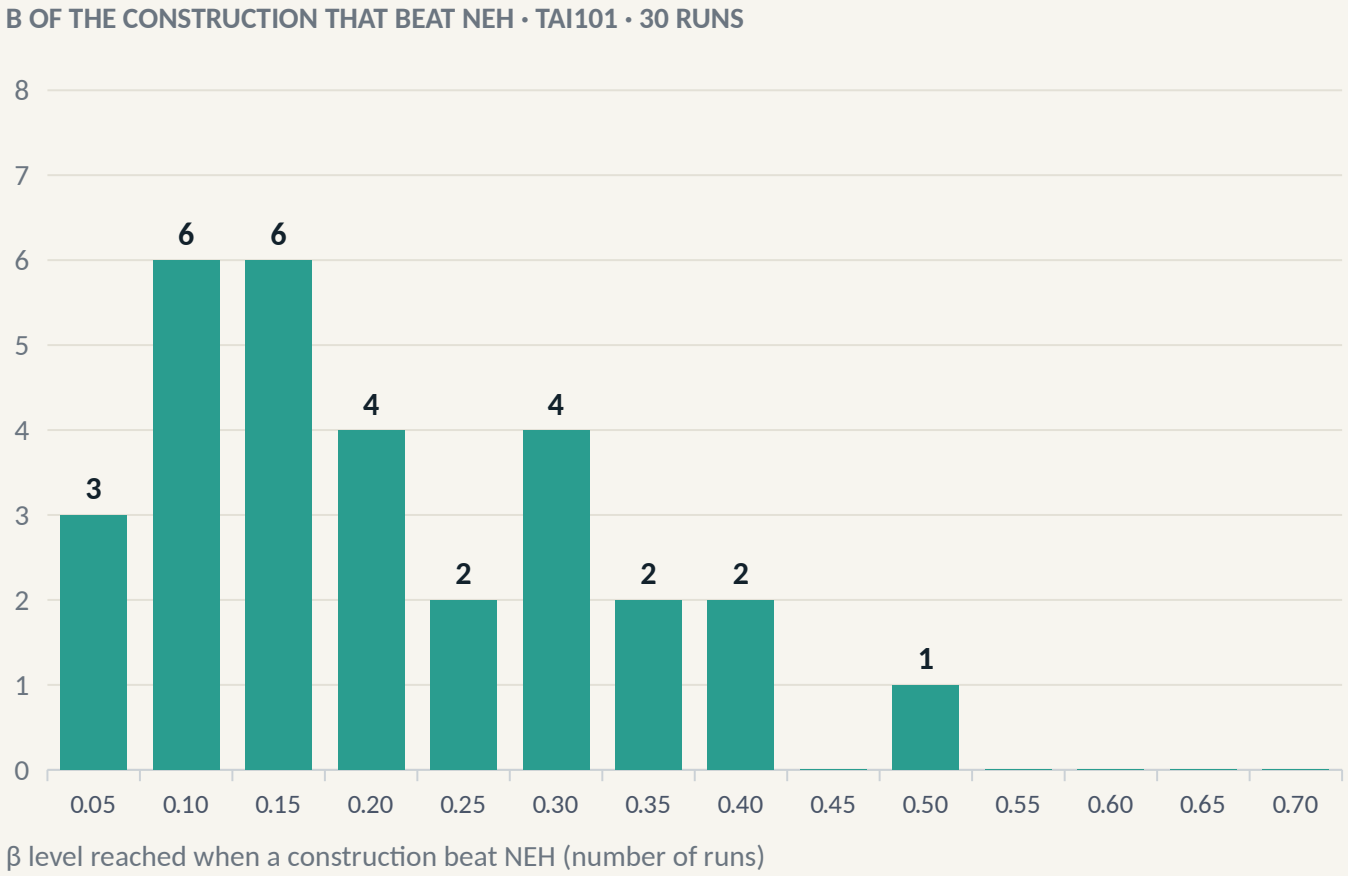
Mean over 30 runs per cell. The ms column is wall-clock on a shared machine, so read it as an order of magnitude.

Plateau. Geo(β) with β from 0.3 to 0.7 gives practically the same GAP, with 1.2 to 7 constructions on average.

Geo-adapt. Same quality, but it costs about as much as Geo(0.1)–(0.2): on tai101, 11.3 constructions against 5.6–6.9 for Geo(0.5)–(0.7).

Triangular and uniform need many more tries: their bias weakens as r grows.

Adaptive β: most runs succeed below β = 0.35



0.05

median β on tai051 and tai117: the first level is usually enough and the adaptation barely acts

0.175

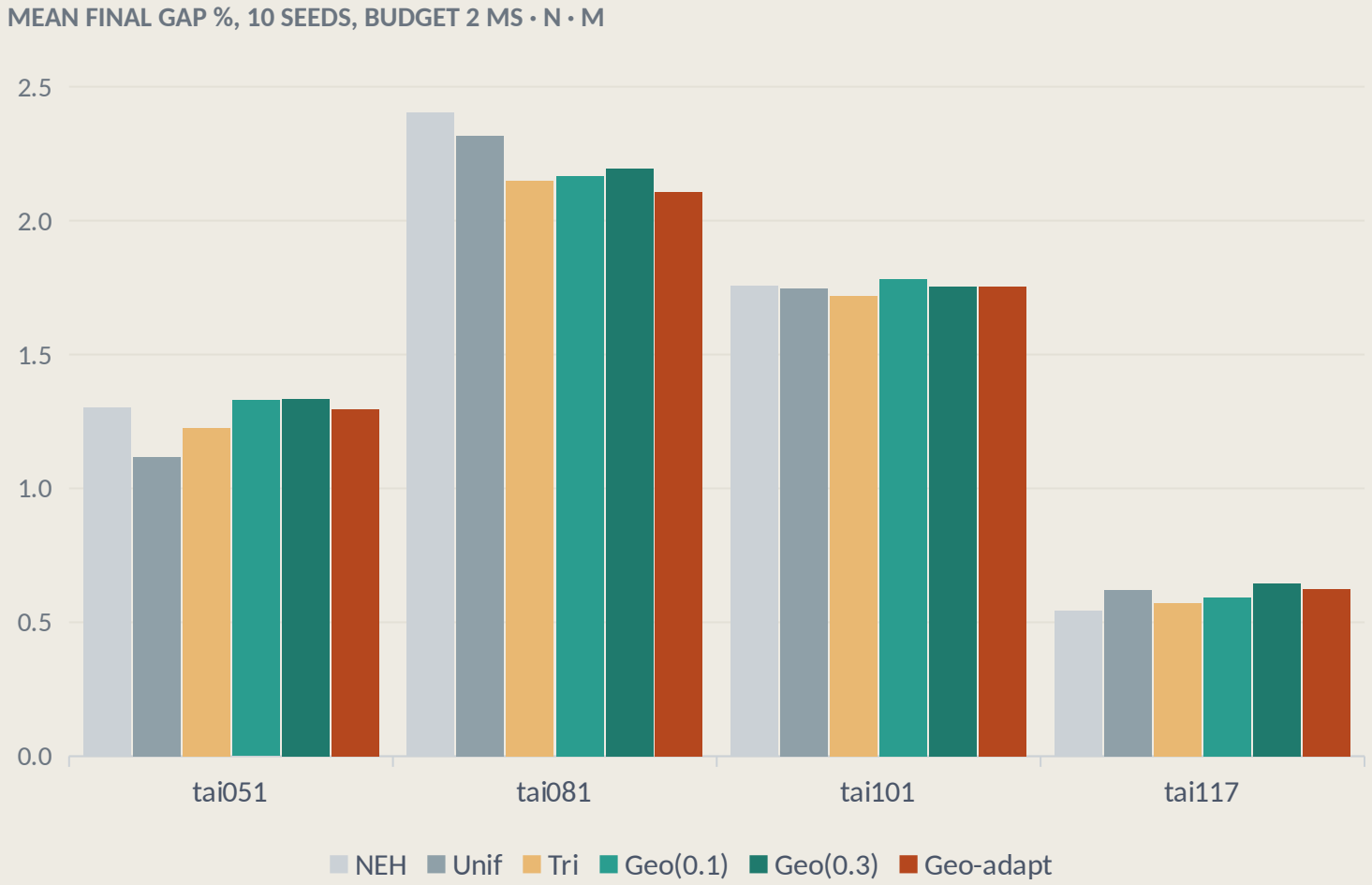
median β on tai101, the most dispersed (range 0.05–0.50); tai081 has one isolated case at 0.60

37 %

of the tai001 runs never beat NEH: with n = 20 the ramp reaches only β = 0.35

Each run restarts at β = 0.05, so the table is the β that worked in each run, not a convergence across runs. A cap at 0.7 was almost never reached.

ILS-ESP: the start barely changes the final result



< 0.3

points between the six starts in every instance, like the spread between seeds (0.09–0.32)

60 / 60

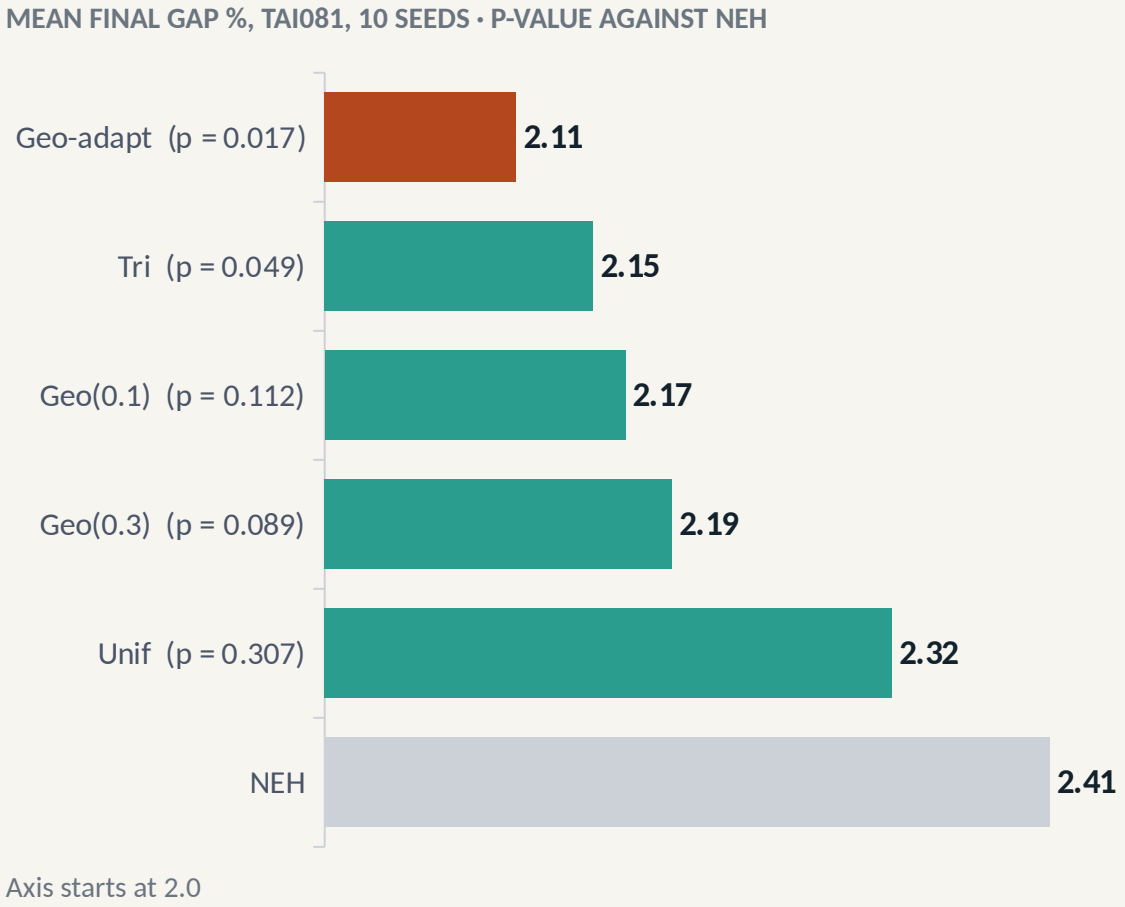
tai001 runs reach GAP 0, whatever the start

–0.30 / +0.08

Geo-adapt against NEH on tai081 (p = 0.017) and tai117 (p = 0.10), uncorrected

tai001 omitted from the chart: GAP 0 for every start. After local search the head start of RandNEH is gone (tai051: NEH 3.94, Tri 4.21, Geo-adapt 3.92).

tai081: randomised starts help



5 / 5

randomised starts end below NEH (2.41), by 0.09–0.30 points

25 tests

uncorrected: none would survive a Bonferroni correction, so this is a hint, not proof

The opposite on tai117: NEH is best on average (0.54 % against 0.57–0.65 %), plausible with only ~250 iterations.

Mean of 10 seeds; p: Mann-Whitney against NEH, uncorrected. Geo-adapt has the lowest spread between seeds here (std 0.14).

TAKEAWAY

Adaptive β works, but a fixed β between 0.3 and 0.7 is cheaper.

RandNEH always beats NEH in the constructive phase; inside ILS-ESP the starting point barely matters.

WHAT WE FOUND

The Fig. 6 RandNEH beats NEH on average in all 5 instances with any distribution: 0.1–0.8 GAP points.

Geo-adapt matches the quality of the fixed-β plateau without choosing β, at a higher cost than a high fixed β.

As an ILS-ESP start it does not stand out: −0.30 on tai081 ($p = 0.017$, uncorrected), +0.08 on tai117.

WORTH ASKING ABOUT

Small instances: with $n = 20$ the ramp is too slow, and 63 % of runs beat NEH against 87–97 % for Geo(0.3)–(0.7).

Noise: the mean final GAP moves ≈ 0.1 with the seed set, and $p < 0.05$ with 25 uncorrected tests is only a hint.

NEXT STEPS

Trying an implementation where the loop doesn’t stop at the first RandNEH successful improvement: are new iterations worth the cost?

Thank you — questions?

BACKUP

Experiment A: mean GAP % and runs that beat NEH

Distribution	tai001 (GAP · %beat)	tai051 (GAP · %beat)	tai081 (GAP · %beat)	tai101 (GAP · %beat)	tai117 (GAP · %beat)
NEH	0.63	6.14	6.00	3.91	1.55
Tri	0.37 · 57 %	5.61 · 100 %	5.75 · 100 %	3.77 · 100 %	1.45 · 100 %
Geo(0.05)	0.43 · 40 %	5.51 · 100 %	5.74 · 100 %	3.76 · 100 %	1.44 · 100 %
Geo(0.1)	0.39 · 53 %	5.66 · 100 %	5.61 · 100 %	3.71 · 100 %	1.40 · 100 %
Geo(0.2)	0.28 · 70 %	5.47 · 100 %	5.72 · 100 %	3.72 · 100 %	1.40 · 100 %
Geo(0.3)	0.23 · 87 %	5.55 · 100 %	5.65 · 100 %	3.76 · 100 %	1.40 · 100 %
Geo(0.5)	0.21 · 97 %	5.44 · 100 %	5.68 · 100 %	3.75 · 100 %	1.39 · 100 %
Geo(0.7)	0.10 · 97 %	5.40 · 100 %	5.74 · 100 %	3.71 · 100 %	1.41 · 100 %
Unif	0.51 · 33 %	5.75 · 100 %	5.87 · 67 %	3.80 · 80 %	1.49 · 97 %
Geo-adapt	**0.36 · 63 %**	**5.58 · 100 %**	**5.68 · 100 %**	**3.73 · 100 %**	**1.41 · 100 %**

30 runs per cell. NEH is a single deterministic solution. Source: results/expA_randneh_resumen.csv.

BACKUP

Experiment B: final GAP % and p-value against NEH

Start	tai001 (mean ± std · p)	tai051 (mean ± std · p)	tai081 (mean ± std · p)	tai101 (mean ± std · p)	tai117 (mean ± std · p)
NEH	0.00 ± 0.00	1.30 ± 0.09	2.40 ± 0.32	1.76 ± 0.15	0.54 ± 0.13
Unif	0.00 ± 0.00	1.12 ± 0.24 · 0.13	2.32 ± 0.23 · 0.31	1.75 ± 0.15 · 0.68	0.62 ± 0.11 · 0.07
Tri	0.00 ± 0.00	1.22 ± 0.22 · 0.35	2.15 ± 0.25 · 0.05	1.72 ± 0.14 · 0.45	0.57 ± 0.11 · 0.28
Geo(0.1)	0.00 ± 0.00	1.33 ± 0.10 · 0.59	2.17 ± 0.31 · 0.11	1.78 ± 0.14 · 0.94	0.59 ± 0.13 · 0.18
Geo(0.3)	0.00 ± 0.00	1.33 ± 0.14 · 0.75	2.19 ± 0.22 · 0.09	1.75 ± 0.13 · 0.91	0.64 ± 0.10 · 0.06
Geo-adapt	**0.00 ± 0.00**	**1.29 ± 0.15 · 0.78**	**2.11 ± 0.14 · 0.02**	**1.75 ± 0.13 · 0.65**	**0.62 ± 0.11 · 0.10**

Best of 10 runs with Tri (equivalent to 10 parallel agents): tai001 0.00 · tai051 0.65 · tai081 1.72 · tai101 1.43 · tai117 0.45. The paper uses 10 ms·n·m and averages 10 instances per set, so this is not a direct comparison.

10 seeds per cell, budget 2 ms · n · m. p: Mann-Whitney, two-sided, uncorrected. Source: results/expB_ilsesp_resumen.csv and expB_tests_vs_NEH.csv.